Lab 1 – Smart City

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CS411W

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1 Introduction

Cities are getting bigger and more crowded every year. The United Nations says that by 2050, around 6.5 billion people will be living in cities. Because of this, people deal with more traffic, less space, and slower communication between city systems. The way most cities are built can't handle this kind of growth, so things like traffic, pollution, and poor resource management keep getting worse.

Smart City is our solution to this problem. It's a community based app that helps people move around the city faster and stay informed. The app gives route suggestions and shares useful updates. The main goal is to reduce congestion and make it easier for residents to stay connected. Smart City helps people and technology work together so cities can be more organized, efficient, and enjoyable to live in.

2 Product Description

Smart City is an app that gives people live updates about what's happening around them. It shows real time traffic and route suggestions based on data from users and sensors.

Overall, Smart City makes city life smoother by combining user data and smart design. It's helpful for anyone, whether you're driving, managing a business, or planning for a city because it uses information from the people who live there to make daily life a little easier.

2.1 Key Product Features and Capabilities

- Key product features include an engaging user interface with community dashboard and navigation
- Smart City looks to solve the problem of rapid urbanization and overcrowding that has been trending globally.
- By implementing a software based solution, we aim to provide citizens with the tools they need to navigate their community without any hassle. By providing a user engaging experience, our application will be able to gather user data in a secure and private manner.

2.2 Major Components (Hardware/Software)

- Hardware needed to support the solution includes a computer to access the web application in charge of handling the data, a smartphone (iOS or Android) for users to access the application for navigation and community hub and Internet connectivity is needed to engage with the application.
- The user application and data platform modules will be developed using the React Native Framework.

3 Identification of Case Study

3.1 Who It's For

- Local city residents and commuters who want traffic, event, and safety updates
- Community members who share or check local updates

3.2 Why

- Make commuting easier and reduce traffic issues
- Help people stay informed about what's happening around them
- Combines navigation, community input, and APIs all into 1 app

3.3 Case Study Group

- Small group of ODU students (5-10) and nearby residents that commute around campus and surrounding area
- They'll use the prototype and give feedback

3.4 Potential Future Uses

- Broader city population (Norfolk, Hampton Roads)
- Visitors or tourists using the app for navigation and local updates

4 Glossary

Internet of Things (IoT): systems that combine sensing, computing, communication, and actuation to connect physical/cyber objects

Application Programming Interface (API): a set of features and rules that lets software interact with other software (a "contract" between programs)

Crowdsourced Reporting: gathering information by soliciting contributions from a large group of people, typically via the internet

Privacy-by-Design ("Privacy-First Design"): building systems so that, by design and by default, only the personal data necessary for a purpose are processed

Open Data (Open Data Access): data anyone can freely use, re-use, and redistribute, with at most attribution/share-alike requirements

Anomaly Detection: identifying data patterns that deviate from expected behavior

Access Control: controls that enforce authorized access to information and system resources in line with policy

Citizen Feedback Loop: a two way process where institutions act on and report back about citizen input to sustain engagement and trust

5 References

[1] Joyce, H. (2025, March 26). Urban overcrowding: A major driver of pollution crises. ShunWaste. https://shunwaste.com/article/how-is-overcrowding-in-cities-causing-more-pollution

[2] Lee, S. (2025, May 27). Understanding urban overcrowding: The social and economic implications of overcrowding in cities. Number Analytics. https://www.numberanalytics.com/blog/ultimate-guide-to-overcrowding-in-urban-sociology

[3] Weinberg, N. (2016, August 11). How can cities be preemptive and effective in preventing overcrowding? Data-Smart City Solutions. https://datasmart.hks.harvard.edu/news/article/howcan-cities-be-preemptive-and-effective-inpreventing-overcrowding-888

[4] Luoma, K. (2016, November 21). How does population growth impact rapid urbanization? Population Education. https://populationeducation.org/how-does-population-growthimpactrapid-urbanization/

[5] Smith, H. (2021, June 18). Post-COVID L.A. traffic analysis: Has rush hour changed? Los Angeles Times. https://www.latimes.com/california/story/2021-06-18/post-covid-l-atrafficanalysis-has-rush-hour-changed

[6] King, G. (n.d.). Data sharing: The next step in municipal collaboration. Data-Command. https://www.data-command.com/the-cloud/data-sharing-the-next-step-in-municipalcollaboration/

[7] Zurich Insurance Group. (2023, April 13). The risks of rapid urbanization in developing countries. https://www.zurich.com/knowledge/topics/global-risks/the-risks-ofrapidurbanization-in-developing-countries

[8] Poindexter, D. (2018, January 24). The effects of overcrowding and the behavioral sink theory. The Borgen Project. https://borgenproject.org/theory-behavioral-sink/

[9] Gorzelany, J. (2024, August 30). Here's where commuters lose the most time and money sitting in traffic, study shows. Forbes. https://www.forbes.com/sites/jimgorzelany/2024/08/30/heres-where-commuters-lose-the-mosttime-and-money-sitting-in-traffic-study-shows/